

Answer Key & Teacher Notes

Session 1 – Understanding Unit Rates • Session 2 – Comparing Unit Prices (Best Buy)

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Understanding Unit Rates

TALK PROMPTS

- What is special about a unit rate? How does writing the rate per 1 unit help you compare amounts?
- How does knowing the miles for one hour let you predict five hours (Problem 2)?

PARTNER TASK

Printer Rate. In Problem 8, one partner finds pages per minute ($288 \div 8 = 36$); the other predicts pages in 15 minutes (540). Check by dividing back: $540 \div 15 = 36$.

WATCH FOR

- Multiplying instead of dividing to find a unit rate (45 miles in 3 hours written as 135).
- Dividing in the wrong order, so the rate is for the wrong unit ($7 \div 84$ instead of $84 \div 7$ for words per minute).
- Leaving off or mixing up the units in the answer.

SESSION 2 • TEACHER NOTES

Comparing Unit Prices (Best Buy)

TALK PROMPTS

- Why can't you just say \$6.25 is cheaper than \$35.00 without doing any math?
- Both packs in Problem 4 cost more than a dollar per notebook. Which pack's buyer pays less for each notebook, and how do you know?

PARTNER TASK

Granola Bars. In Problem 8, one partner finds Box X's price per bar (\$0.75); the other finds Box Y's (\$0.70). Decide which is the better buy and by how much (\$0.05 per bar).

WATCH FOR

- Choosing the package with the lower total price instead of the lower unit price.
- Assuming the bigger package is always the better buy.
- Dividing the number of units by the price instead of the price by the number of units.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.RP.A.2	1	2	1	1	2	2	2	2	3	1	1
S1 • Form B	6.RP.A.2	1	2	1	1	2	2	2	2	3	1	1
S2 • Form A	6.RP.A.2	1	2	2	2	2	2	2	2	3	1	1
S2 • Form B	6.RP.A.2	1	2	2	2	2	2	2	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

rate: A comparison of two amounts with **different** units.

unit rate: The amount for exactly **1** unit.

per: Means “for **each**” one.

constant rate: The amount stays the **same** every time.

- To find a unit rate, **divide** the first amount by the second.
- Multiply the unit rate by any number of units to **predict**.
- Always label the **units**: miles per hour, dollars per pound.

Example: Divide to find 1 minute: $90 \div 6 = 15$ pages per minute. Multiply for 7 minutes: $15 \times 7 = 105$. Check: $15 \times 6 = 90$. Answer: **105** pages in 7 minutes

YOU TRY

1. a. 12 words per minute b. \$7 per hour
2. $150 \div 3 = 50$ mph; $50 \times 5 = 250$ miles

EXIT TICKET

1. a. 45 miles per hour b. \$8 each
2. $315 \div 7 = 45$ per minute; $45 \times 20 = 900$ bottles
3. **B** (60 miles per hour)

PRACTICE FORM A

1. a. 12 words per minute b. \$7 per hour c. 27 miles per gallon d. 11 steps per second e. 16 beads each f. 23 lb per bed
2. a. $150 \div 2 = 75$ miles per hour b. $75 \times 5 = 375$ miles
3. $\$45 \div 3 = \15 per lawn
4. $\$7.50 \div 6 = \1.25 per bottle
5. a. $240 \div 8 = 30$ miles per gallon b. $360 \div 30 = 12$ gallons
6. a. $12 \div 2 = 6$ miles per hour b. $18 \div 6 = 3$ hours
7. a. A: $\$54 \div 6 = \9 per hour; B: $\$85 \div 10 = \8.50 per hour b. Job A, by \$0.50 per hour c. $14 \times \$8.50 = \119
8. a. $288 \div 8 = 36$ pages per minute b. $36 \times 15 = 540$ pages
9. A unit rate needs division, not multiplication. 135 mph is impossible for a bike. $45 \div 3 = 15$ miles per hour.
10. **B** (60 miles per hour)
11. **A** (\$0.50 per apple)

Self-check word DIVIDES

PRACTICE FORM B

1. a. 13 words per minute b. \$9 per hour c. 28 miles per gallon d. 12 steps per second e. 14 beads each f. 26 lb per bed
2. a. $195 \div 3 = 65$ miles per hour b. $65 \times 4 = 260$ miles
3. $\$52 \div 4 = \13 per lawn
4. $\$9.00 \div 12 = \0.75 per bottle
5. a. $288 \div 9 = 32$ miles per gallon b. $352 \div 32 = 11$ gallons
6. a. $15 \div 3 = 5$ miles per hour b. $20 \div 5 = 4$ hours
7. a. A: $\$60 \div 8 = \7.50 per hour; B: $\$72 \div 9 = \8 per hour b. Job B, by \$0.50 per hour c. $12 \times \$8 = \96
8. a. $312 \div 8 = 39$ pages per minute b. $39 \times 12 = 468$ pages
9. A unit rate needs division, not multiplication. 96 mph is impossible for a runner. $24 \div 4 = 6$ miles per hour.
10. **B** (70 miles per hour)
11. **C** (\$0.40 per pen)

Self-check word COMPARE

GUIDED NOTES

unit price: The cost of exactly **one** item or unit.

better buy: The choice with the **lower** unit price.

total price: What the **whole** package costs.

- Unit price = total cost $\div$ number of **units**.
- Find the unit price of **each** package, then compare.
- The **lower** unit price is the better buy.

Example: A: $\$7.50 \div 5 = \1.50 per notebook. B: $\$10.40 \div 8 = \1.30 per notebook. $\$1.30 < \1.50 , so Store **B** is cheaper. Answer: Store B saves **\$0.20** per notebook.

YOU TRY

1. a. \$0.75 each b. \$0.25 each
2. $\$2.40 \div 12 = \0.20 per oz; $\$3.60 \div 20 = \0.18 per oz; the 20-oz bottle

EXIT TICKET

1. a. \$1.50 per lb b. \$0.50 each
2. A: \$0.70 per bar; B: \$0.65 per bar; Pack B
3. **B** (The price per ounce of each jar)

PRACTICE FORM A

1. a. \$0.75 each b. \$2.25 per yd c. \$0.25 each d. \$2.20 per lb e. \$0.15 per oz f. \$0.50 each
2. a. $\$35.00 \div 7 = \5 per day b. Option B c. $\$6.25 - \$5.00 = \$1.25$ per day
3. a. $\$4.00 \div 16 = \0.25 b. $\$4.80 \div 24 = \0.20 c. Brand B
4. Pack 1: \$1.50; Pack 2: \$1.40. Pack 2
5. Streamify \$9; TuneCloud \$8. The lower fee is \$8 per month (TuneCloud).
6. 3 pairs: \$3.50 each; 5 pairs: \$3.25 each; savings \$0.25 per pair
7. a. \$0.09 per oz and \$0.085 per oz b. $4 \times \$2.88 = \11.52 ; $2 \times \$5.44 = \10.88
8. a. $\$6.00 \div 8 = \0.75 b. $\$8.40 \div 12 = \0.70 c. Box Y
9. She compared total prices, not unit prices. 12 oz: \$0.20 per oz; 20 oz: \$0.18 per oz. The 20-oz bottle is the better buy.
10. **A** (\$0.60)
11. **B** (4 lb for \$9)

Self-check word BARGAIN

PRACTICE FORM B

1. a. \$0.70 each b. \$1.60 per yd c. \$0.30 each d. \$1.30 per lb e. \$0.25 per oz f. \$0.75 each
2. a. $\$24.00 \div 5 = \4.80 per day b. Option B c. $\$5.50 - \$4.80 = \$0.70$ per day
3. a. $\$5.00 \div 20 = \0.25 b. $\$7.04 \div 32 = \0.22 c. Brand B
4. Pack 1: \$1.25; Pack 2: \$1.35. Pack 1
5. Streamify \$7.50; TuneCloud \$6. The lower fee is \$6 per month (TuneCloud).
6. 4 pairs: \$3.25 each; 6 pairs: \$3.05 each; savings \$0.20 per pair
7. a. \$0.09 per oz and \$0.085 per oz b. $4 \times \$2.16 = \8.64 ; $2 \times \$4.08 = \8.16
8. a. $\$6.50 \div 10 = \0.65 b. $\$9.30 \div 15 = \0.62 c. Box Y
9. He compared total prices, not unit prices. 16 oz: \$0.20 per oz; 24 oz: \$0.17 per oz. The 24-oz bag is the better buy.
10. **A** (\$0.80)
11. **A** (5 lb for \$11)

Self-check word SAVINGS

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • UNDERSTANDING UNIT RATES

1. a. 45 mi per hour b. \$8 per ticket c. 70 words per minute d. 12 apples per bag
2. 1 h: \$12; 2 h: \$24; 4 h: \$48. Unit rate: \$12 per hour ($36 \div 3 = 12$).
3. $156 \div 6 = 26$ smoothies per hour; $26 \times 9 = 234$ smoothies in 9 hours
4. a. $12 \div 4 = 3$ cups per loaf b. $4 \div 12 = 1/3$ loaf per cup
5. Ben divided in the wrong order; 0.02 is minutes per word. Words per minute = $150 \div 3 = 50$ words per minute.

SHEET 2 • COMPARING UNIT PRICES (BEST BUY)

1. a. \$1.50 per lb b. \$0.50 per box
2. Small: $\$3.60 \div 12 = \0.30 per oz; Large: $\$5.04 \div 18 = \0.28 per oz. The large box is the better buy.
3. a. $\$46 \div 2 = \23 per gallon b. $\$107.50 \div 5 = \21.50 per gallon c. $\$23 - \$21.50 = \$1.50$ per gallon
4. a. A: $\$5.00 \div 4 = \1.25 ; B: $\$7.80 \div 6 = \1.30 b. Store A ($\$1.25 < \1.30)
5. No. 32-oz jar: $\$5.12 \div 32 = \0.16 per oz; 16-oz jar: $\$2.40 \div 16 = \0.15 per oz. The smaller jar is the better buy.
6. **B** (5 lb for \$7.00)
7. **A** (\$0.45)